

AZURE VIRTUAL DESKTOP POOLED HOST POOL FOR AN ENGINEERING LAB

Comprehensive Project Implementation & Capacity Evaluation Report

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Academic Institution: **K L University** | Subscription: **Azure for Students**

1. Introduction & Academic Objectives

This project develops an enterprise pooled desktop virtualization platform on Microsoft Azure designed to host engineering software for a student body of approximately 50 students. The project aligns academic virtualization requirements with industry best practices, addressing multi-session compute sizing, FSLogix profile storage performance, and strict student subscription cost governance.

2. Existing Environment & Non-Destructive Integration

Prior to implementation, the existing Azure infrastructure was audited and preserved without recreation:

Resource Name	Resource Type	Live Azure Configuration	Status
rg-avd-engineering-lab	Resource Group	indiasouthcentral	Preserved
vnet-avd-lab	Virtual Network	10.0.0.0/16 address space	Preserved
snet-avd	Subnet	10.0.0.0/24 address prefix	Preserved
stavdprofiles260916	Storage Account	FileStorage, StandardV2_LRS, TLS 1.2	Preserved
fslogix	Azure File Share	128 GiB, 1026 IOPS, 63 MiB/s SMB share	Preserved
AADKERB App Registration	Entra Application	Client ID: 2224dfcc-7a68-425f-a668-4fcfb5b4ceb0	Preserved

3. Implementation Phases & Faculty Tasks Alignment

- **Phase 1 & 2 (Resource Providers):** Registered Microsoft.Compute and Microsoft.DesktopVirtualization at subscription scope.
- **Phase 3 (Network Security - Task 4):** Deployed NSG nsg-avd-lab with explicit outbound rules for AVD control plane (443), Azure Files (445/443), and KMS (1688). Associated with snet-avd.
- **Phase 4 (AVD Control Plane):** Deployed Host Pool hp-avd-engineering-lab (BreadthFirst, Max 5 sessions), Desktop App Group dag-avd-engineering-lab, and Workspace ws-avd-engineering-lab anchored in centralindia.
- **Phase 5 (Entra RBAC - Task 2):** Created security group AVD-Engineering-Lab-Students, enrolled student account 2400031215@kluniversity.in, and assigned Desktop Virtualization User role on the DAG.
- **Phase 6 (Resource Lock - Task 3):** Applied CanNotDelete lock lock-storage-prevent-delete on stavdprofiles260916.

4. Compute Quota Analysis & Session Host Sizing

A live audit of subscription core limits in indiasouthcentral revealed that the Standard Dv5 family has **0 quota** under Azure for Students. Active quota exists for the **Standard BS Family (Limit: 4 vCPUs)** and **Standard Dv4 Family (Limit: 4 vCPUs)**. The project selects **Standard_B4ms** (4 vCPUs, 16 GiB RAM, ~\$0.166/hr) for economical initial testing, delivering an estimated 2-hour test cost of ~\$0.33.

5. FSLogix Profile Storage & Bottleneck Evaluation

User profiles are stored in dynamic VHDX containers on the fslogix file share. With 1026 provisioned IOPS and 5000 burst IOPS, the storage tier provides approximately 20.5 IOPS per student under full 50-user concurrency, comfortably exceeding the recommended 10–15 IOPS steady-state threshold for standard engineering profiles. Login storm mitigation is further bolstered by burst IOPS credits during initial session mounts.

6. Conclusion & Capacity Sizing for 50 Students

The architecture establishes that supporting 50 enrolled students does not necessitate 50 concurrent virtual machines. With multi-session capability supporting 4–5 concurrent engineering students per Standard_B4ms host, the entire laboratory cohort can be reliably served by **10 to 12 session hosts**, yielding significant infrastructure cost optimization compared to dedicated physical workstations.